BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

PHYSICS (10th) — (Fresh/Re-appear) — (Version-A) — SECTION –A

Time Allowed: 20 Min | **Maximum Marks:** 12 | **Paper Code:** BN-X-XXIII-I

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. If iron bob of a pendulum is replaced with lead bob of the same size, its time period will;

(A) Decrease
(B) Increase
(C) Remains the same
(D) Zero

Q2. The loudness of a sound is most closely related to its;

(A) Frequency
(B) Amplitude
(C) Period
(D) Wavelength

Q3. The focal length of a concave spherical mirror is equal to;

(A) R
(B) 2R
(C) 1/4 R
(D) 1/2 R

Q4. Which of the following is an example of longitudinal waves?

(A) Water waves
(B) Sound waves
(C) Light waves
(D) Microwaves

Q5. The capacitance is defined as;

(A) W/Q
(B) Q/Q
(C) V/Q
(D) Q/V

Q6. The SI unit of resistance is;

(A) Ohm
(B) Ampere
(C) Volt
(D) Watt

Q7. The principal of AC generator is;

(A) Magnetic field
(B) Electrostatic induction
(C) Electromagnetic induction
(D) Mutual induction

Q8. If the turns ratio is 5, the secondary voltage is greater than the primary voltage by a factor of;

(A) 2.5
(B) 5
(C) 0.5
(D) 10

Q9. The Boolean expression for OR gate is;

(A) X = A with overline
(B) X = AB
(C) X = A+B with overline
(D) X = A+B

Q10. Which of the following information storage device is least used these days?

(A) Floppy
(B) CD
(C) HDD
(D) USB

Q11. What is the number of protons in $_{7}N^{15}$?

(A) 15
(B) 8
(C) 7
(D) 6

Q12. Time rate of flow of charge is called;

(A) Electric potential
(B) Current
(C) Resistance
(D) Capacitance

PHYSICS (10th) — (Fresh/Re-appear) — SECTION –B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 32

> Note: Note: Attempt section B & C accordingly.

Q2. (Marks 32) Q.No 2. Attempt any EIGHT parts of the following. All parts carry equal marks.

i. Derive a relation between speed, frequency and wavelength of a wave.

ii. Differentiate between infrasonics and ultrasonics.

iii. What is the frequency of sound with wavelength 0.25m in air with temperature of 32°C?

iv. Describe power of lens and its unit.

v. Explain why is it dangerous to turn ON a light bulb when you are in a bath tub?

vi. Why are lightening rods normally at a higher elevation than the buildings they protect?

vii. Describe the importance of fuse and circuit breaker.

viii. Why does a compass needle points north?

ix. What is meant by thermionic emission?

x. Differentiate between hardware and software.

xi. If nuclear radiation is harmful, how it can be used for treatment of diseases?

SECTION –C

Maximum Marks: 21

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) What is meant by refraction of light? Explain with diagrams.

b) The distance between successive crests in a series of water waves is 4m and the crests travel 9m in 4.5s. Calculate the frequency of the waves.

Q4.

a) State and explain the phenomenon of mutual induction.

b) What is the wavelength of sound in dry air if its frequency is 590 Hz on a hot day with temperature of 40°C?

Q5.

a) What is Ohm's law? Describe its limitations.

b) A 1.5m long wire carries a current of 5A at right angle to a uniform magnetic field of 0.04T. Determine the force exerted on the wire.

Q6.

a) Explain alpha decay and beta decay. Give one example of each.

b) The potential difference across the plates of a capacitor is 500V. The charge on each plate is 0.02C. What is the capacitance of the capacitor?